

VIRAL VISHAL DALAL

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Education

New York University

Masters of Science in Computer Engineering - 4 GPA

September 2025 - May 2027

New York, New York

University of Mumbai

Bachelor of Technology in Information Technology with Honors in DevOps - 3.6 GPA

December 2021 - May 2025

Mumbai, India

Coursework: Data Structures & Algorithms, Distributed Systems, Software Engineering, Artificial Intelligence, MLOps

Technical Skills

Languages & Frameworks: Python, C++, C#, Java, HTML, CSS, JavaScript, SQL, Go, React.js, Node.js, Next.js, R
ML: Pandas, NumPy, Matplotlib, NLP, Scikit-learn, PyTorch, TensorFlow, LangChain, Pinecone, Airflow, LangGraph, Flask
Tools & Databases: AWS, Git, Azure, Kubernetes, Docker, MongoDB, PostgreSQL, Mysql, Bedrock, FastAPI, Linux, MCP

Experiences

Adobe

Software Engineer Intern

May 2026 – Aug 2026

San Jose, CA

- Working on the Adobe Journey Optimizer Platform.

SimPPL

Founding AI Engineer Intern

June 2024 – June 2025

Remote, New York

- Founded an **AI-powered RAG** platform via **Python** for grant writing, **increasing** application **throughput** by **30%**.
- Built automated document ingestion, embedding-based retrieval via **FAISS** & **Pinecone**, memory-aware conversations, **DSPy**-based query optimization, & a web-search AI agent for grantor-aligned context enrichment.
- Built an AI-powered audience **analytics** platform MVP leveraging **Google BigQuery** for data storage & querying; piloted with **Mozilla/Google** grants, enabling **New York Public Radio** segmentation & engagement for **20K+** users.
- Developed a **multi-agent** debate system using **LangChain** for content moderation on **'X'**, with LLM agents evaluating content safety for **8k+** tweets collected using **AWS EC2**. Research accepted at the **Stanford's T&S** conference. [Paper](#)

IIT Patna

Machine Learning Research Intern

June 2024 – March 2025

Remote, India

- Developed a **multi agent orchestration**, **Agentic AI** framework using **CrewAI** & **Autogen** to diagnose critical failure points in LLM-generated peer reviews. Validated with **pytest** and assessed model evaluation metrics. [Paper](#)
- Implemented **GraphRAG** using **Neo4j** to assess peer review consistency, & achieved **93%** consistency by leveraging knowledge graphs at word, sentence, and aspect levels to improve transparency in peer reviews. [Paper](#)
- Developed **Chain-of-Thought** reasoning framework using LLM models for automated peer review classification, into 'exhaustive' or 'trivial' reviews via section & aspect coverage analysis. [Paper](#)

Invexi

Software and AI Engineering Intern

April 2024 – November 2024

Remote, Singapore

- Created a **browser extension** with **TypeScript**, **GoLang**, **PostgreSQL** & containerized using **Docker** to **automate** on Ubuntu based **Linux** servers product data collection & providing AI based analysis, aiding manual market research.
- Engineered an **ETL** analytics pipeline for stock screening, reducing processing time by **15%** while automating data ingestion, transformation, technical indicator computation, and real time AI powered analysis via **RESTful APIs**.

Drishya Solutions

Software Developer Intern

June 2023 – September 2023

Mumbai, India

- Architected a full-stack CRM software for a client; normalized **MySQL** schema and exposed **backend REST APIs** to manage & query **SQL** data, with caching, input validation, and **CI/CD** (build/test/deploy) to deliver a responsive UI.
- Executed **Agile** delivery across the Software Development Life Cycle with rigorous testing & systematic debugging.

Projects

Volunteer Platform - Morgan Stanley Code to Give USA 2026 Winner | LangGraph, AWS Bedrock, Next.js, FastAPI

- Built **Next.js** & **TypeScript** frontend, **FastAPI** backend with **Supabase PostgreSQL** database, **15** tables with Row-Level Security, AI conversational campaign builder, & gamification. Deployed on **Vercel** with **200+** **pytest** tests.
- Designed a 3-node **LangGraph** agentic AI system on **AWS Bedrock** with conditional routing, **LangChain** orchestration of 12 tools, **RAG** style retrieval from **live API**, auto-chaining, **prompt engineering**, **900K+** users.

FitZen - GeeksForGeeks Hackathon Winner | React.js, Node.js, Python, FireBase, MediaPipe, AI, NLP, Tensorflow

- Built a full-stack fitness app with realtime pose detection & rep counting using **MediaPipe**, improving accuracy in form.
- Implemented AI-guided diet & workout recommendation system, a mobile app nutrient scanner via **RESTful APIs**.

Gen-AI Driven Sustainable Packaging Solution | React.js, Node.js, LLMs, PyTorch, CUDA, OpenCV | [Paper](#)

- Developed 2D packaging visuals & dielines with **fine-tuned** Stable Diffusion, converted it into **3D** models via **Pytorch**.
- Added sustainable material suggestions via Mistral-7B with **LoRA/PEFT** (4-bit QLoRA) on T4 **GPU**, tracked experiments with Weights & Biases via MLFlow, deployed on **Hugging Face** Transformers with React & **Node.js**.